Pixel 7a Main Camera: Measured Datasheet (condensed)

Google Pixel 7a, main (1x) camera, from raw DNG files.

Match with the published datasheet: 49 of 49 values match (ALL MATCH).
Unit tests on synthetic data with known answers: (tests not run)

results.json (value, uncertainty):

camera	Pixel 7a
capture_app	Open Camera (Camera2 API, JPEG and DNG)
cfa_pattern	RGGB
raw_width_px	4624
raw_height_px	3472
binned_from_higher_resolution_sensor	True
focal_length_px	[3475.58, 5.94]
focal_length_mm	[5.43, 0.005]
principal_point_x_px	[2238.33, 6.9]
principal_point_y_px	[1736.19, 3.75]
distortion_k1	[0.05448, 0.00482]
distortion_k2	[-0.26879, 0.03272]
distortion_p1	[-0.000257, 0.000325]
distortion_p2	[7.8e-05, 0.00035]
reprojection_rms_px	[1.062, 0.301]
n_calibration_images_used	17
checkerboard_square_mm_measured	[19.2, 0.1]
mtf50_center_cy_per_px	[0.4726, 0.0169]
mtf50_corner_mean_cy_per_px	[0.221, 0.0723]
mtf50_corner_worst_cy_per_px	[0.0867, 0.0931]
conversion_gain_dn_per_e	[0.07024, 0.00107]
read_noise_e	[7.094, 1.104]
dark_current_e_per_s	[0.9572, 0.0165]
iso_of_noise_measurement	45
black_level_dn	[64.009, 0.0009]
white_level_dn	1023
vendor_noiseprofile_present	True
dynamic_range_stops	[10.81, 0.21]
dynamic_range_criterion	EMVA 1288: saturation capacity (white level, 1023 DN) over the absolute sensitivi
delta_e_metric	CIEDE2000
colorchecker_reference_set	ColorChecker24 - After November 2014
delta_e_mean_before_ccm	[10.64, 0.24]
delta_e_mean_after_ccm	[3.77, 0.51]
delta_e_worst_after_ccm	[10.71, 0.4]

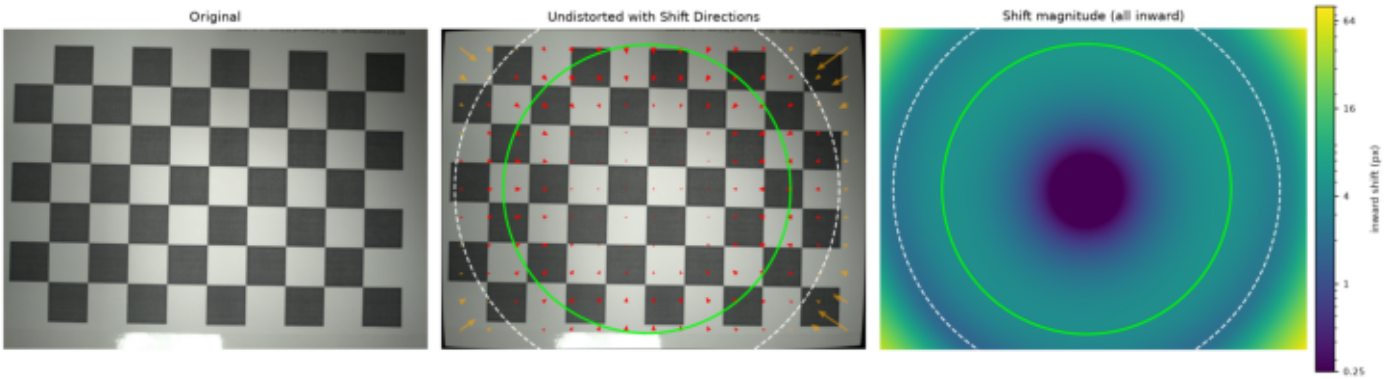
Computed values against the published datasheet

OP	quantity	unit	computed	datasheet	match
OP1	focal length f_x	px	3477.0 ± 8.3	3477.0 ± 8.3	**PASS**
OP1	focal length f_y	px	3474.2 ± 8.5	3474.2 ± 8.5	**PASS**
OP1	focal length, mean	px	3475.6 ± 5.9	3475.6 ± 5.9	**PASS**
OP1	principal point c_x	px	2238.3 ± 6.9	2238.3 ± 6.9	**PASS**
OP1	principal point c_y	px	1736.2 ± 3.8	1736.2 ± 3.8	**PASS**
OP1	radial k1		0.054 ± 0.005	0.054 ± 0.005	**PASS**
OP1	radial k2		-0.27 ± 0.03	-0.27 ± 0.03	**PASS**
OP1	radial k3		0.35 ± 0.07	0.35 ± 0.07	**PASS**
OP1	tangential p1		-0.0003 ± 0.0003	-0.0003 ± 0.0003	**PASS**
OP1	tangential p2		0.0001 ± 0.0004	0.0001 ± 0.0004	**PASS**
OP1	RMS reprojection	px	1.06 ± 0.30	1.06 ± 0.30	**PASS**
OP1	views used		17	17	**PASS**
OP1	output pixel pitch (	um	1.56 ± 0.04	1.56 ± 0.04	**PASS**
OP1	diagonal field of vi	deg	79.5 ± 0.1	79.5 ± 0.1	**PASS**
OP2	MTF50 screen, cen	cy/px	0.473 ± 0.017	0.473 ± 0.017	**PASS**
OP2	MTF50 screen, bott	cy/px	0.25 ± 0.09	0.25 ± 0.09	**PASS**
OP2	MTF50 screen, top	cy/px	0.27 ± 0.09	0.27 ± 0.09	**PASS**
OP2	MTF50 screen, top	cy/px	0.09 ± 0.09	0.09 ± 0.09	**PASS**
OP2	MTF50 screen, bott	cy/px	0.20 ± 0.09	0.20 ± 0.09	**PASS**
OP2	MTF50 screen, corr	cy/px	0.22 ± 0.07	0.22 ± 0.07	**PASS**
OP2	corner mean witho	cy/px	0.25 ± 0.04	0.25 ± 0.04	**PASS**
OP2	worst corner witho	cy/px	0.20 ± 0.04	0.20 ± 0.04	**PASS**
OP2	center MTF50 / Nyq	%	95	95	**PASS**
OP2	modulation at Nyqı		0.47 ± 0.02	0.47 ± 0.02	**PASS**
OP2	MTF50 printed targ	cy/px	0.189 ± 0.005	0.189 ± 0.005	**PASS**
OP2	MTF50 printed targ	cy/px	0.182 ± 0.008	0.182 ± 0.008	**PASS**
OP2	modulation at Nyqı		0.13 ± 0.03	0.13 ± 0.03	**PASS**
OP3	conversion gain K	DN/e-	0.0702 ± 0.0011	0.0702 ± 0.0011	**PASS**
OP3	read noise sigma_d	e-	7.1 ± 1.1	7.1 ± 1.1	**PASS**
OP3	sigma_y.dark	DN	0.543	0.543	**PASS**
OP3	SNR_max		116.8 ± 0.9	116.8 ± 0.9	**PASS**
OP3	abs. sensitivity thr	e-	7.6 ± 1.1	7.6 ± 1.1	**PASS**
OP3	saturation capacity	e-	13650 ± 210	13650 ± 210	**PASS**
OP3	dynamic range (EM	bit	10.8 ± 0.2	10.8 ± 0.2	**PASS**
OP3	dynamic range	dB	65.1	65.1	**PASS**
OP3	photon/read-noise	e-	50 ± 16	50 ± 16	**PASS**
OP3	dark current (G1)	e-/s	0.957 ± 0.016	0.957 ± 0.016	**PASS**
OP3	vendor NoiseProfile	DN/e-	0.0788	0.0788	**PASS**
OP3	K, ordinary least sq	DN/e-	0.0691 ± 0.0002	0.0691 ± 0.0002	**PASS**
OP3	sigma_d, ordinary l	e-	8.75 ± 0.64	8.75 ± 0.64	**PASS**
OP3	K, 25 patches, illun	DN/e-	0.0699 ± 0.0005	0.0699 ± 0.0005	**PASS**
OP3	single-frame K (wr	DN/e-	0.1178	0.1178	**PASS**
OP4	mean dE00, white		10.64 ± 0.24	10.64 ± 0.24	**PASS**
OP4	worst dE00, white		18.05 ± 0.12	18.05 ± 0.12	**PASS**
OP4	mean dE00, 3x3 CC		3.77 ± 0.51	3.77 ± 0.51	**PASS**

Computed values against the published datasheet (continued)

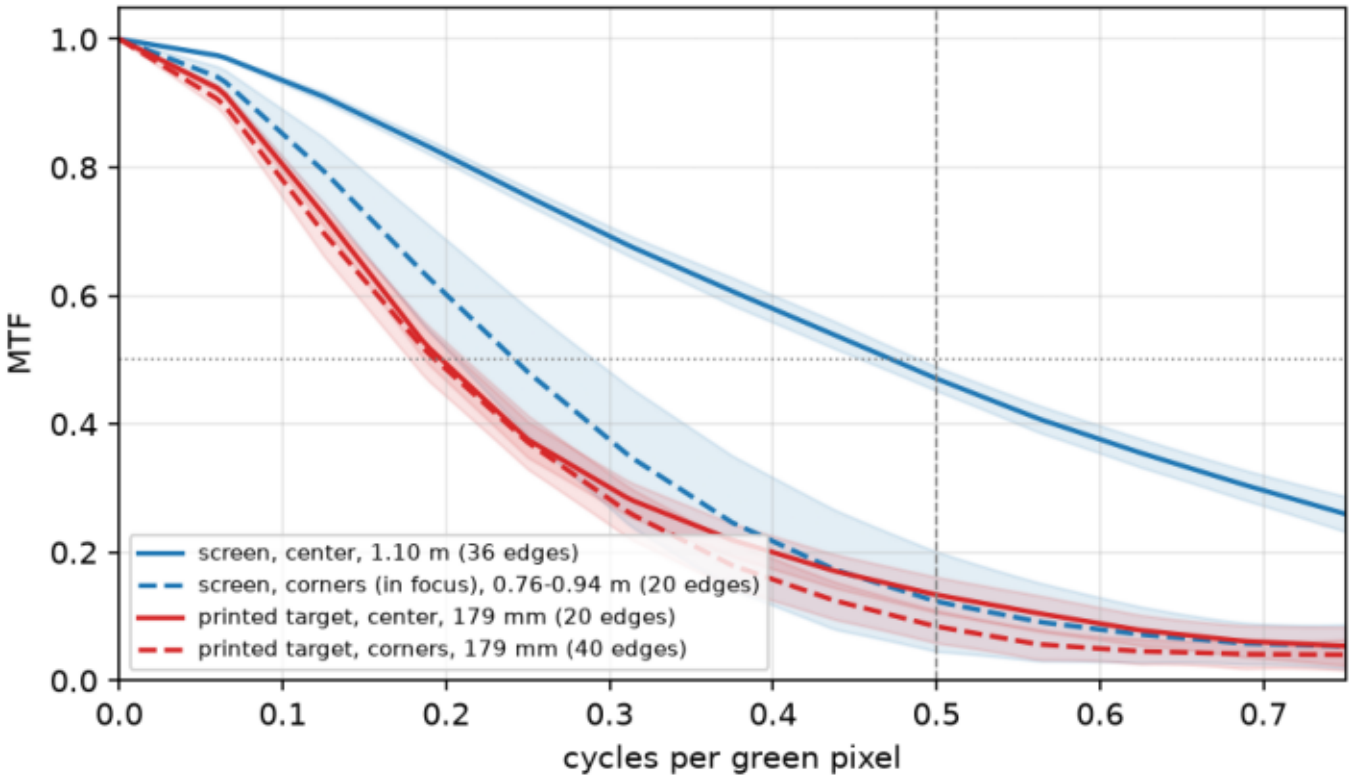
OP	quantity	unit	computed	datasheet	match
OP4	same CCM, LED lig		4.23 ± 0.22	4.23 ± 0.22	**PASS**
OP4	same CCM, outdoo		8.17 ± 0.47	8.17 ± 0.47	**PASS**
OP4	matrix columns ag	deg	2.65	2.65	**PASS**

OP1 Geometry



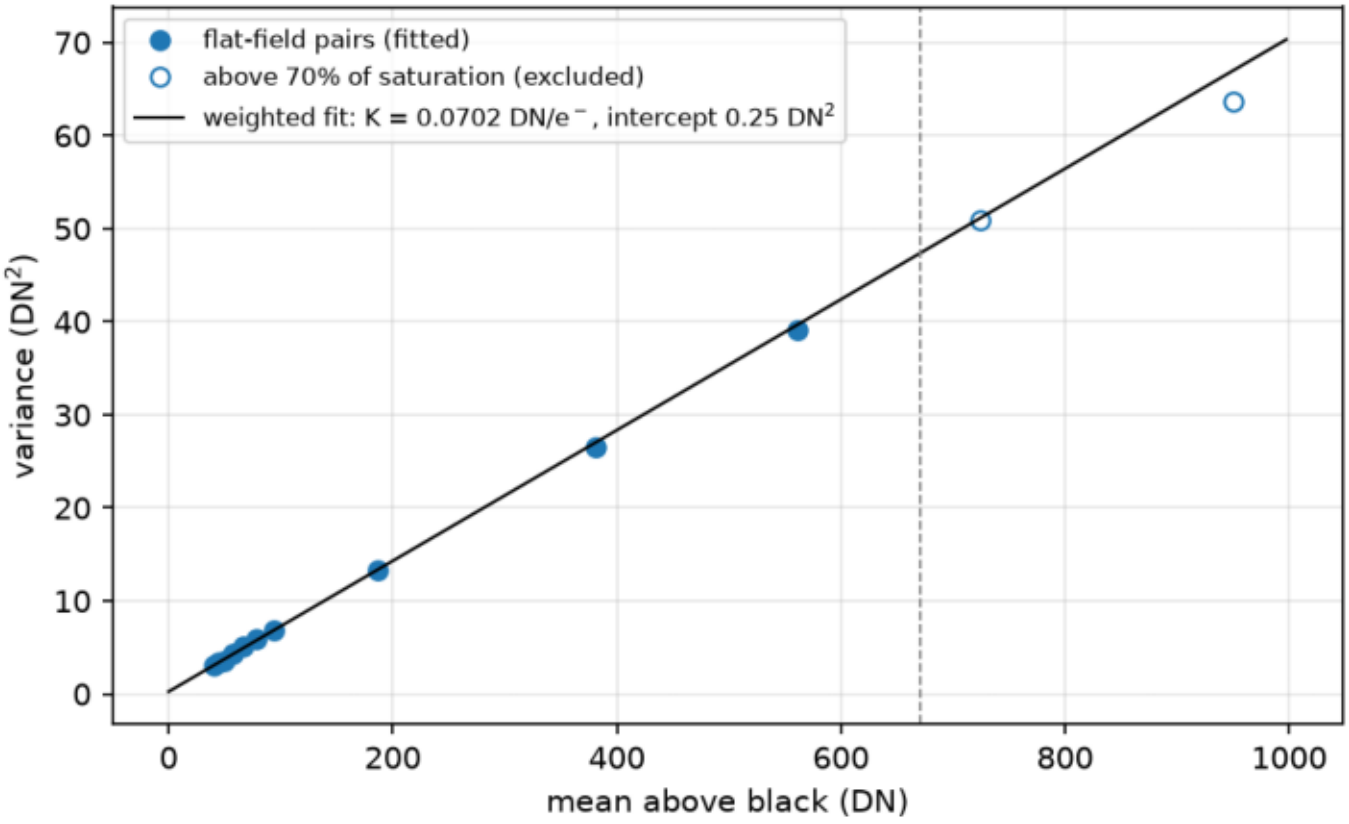
Left: original raw capture. Center: undistorted, with arrows (x15) from each grid point's original position to its undistorted one; orange arrows lie beyond the calibrated field. Right: size of the inward shift. Green circle: field where the k1 and k2 terms are equal; dashed: the farthest calibration corner, beyond which the model is extrapolated.

OP2 Spatial resolution



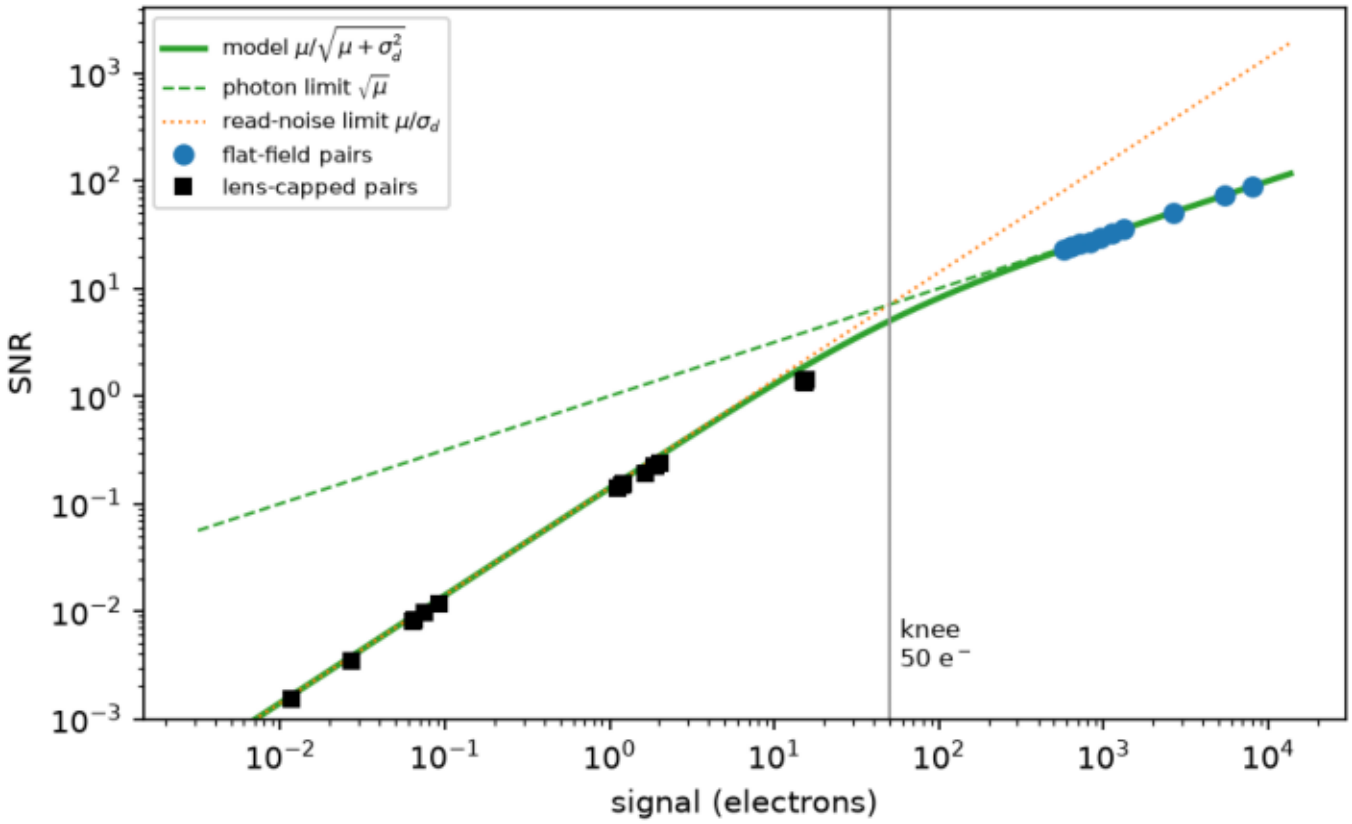
Mean MTF (band: +/-1 std. dev. over edges), same +/-8 px LSF window for all. Blue: slanted edge on a laptop screen (the camera without a print). Red: printed target at 179 mm.

OP3 Noise: photon transfer curve



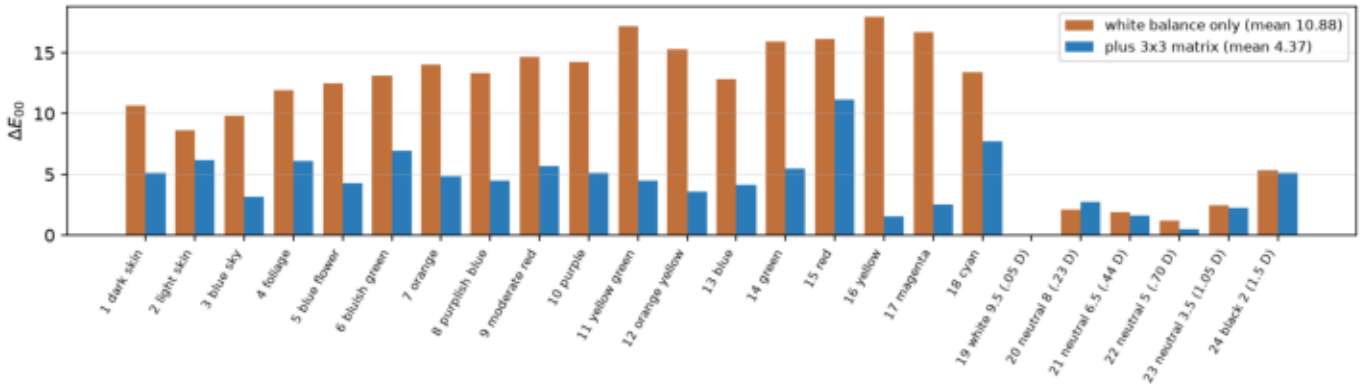
Half the pair-difference variance against the mean above black, central 128 x 128 G1 patch; open points (above 70% of saturation) excluded. Slope K, intercept K^2 sigma_d^2.

OP3 Noise: signal-to-noise ratio



SNR against signal with the model from K and sigma_d; squares: lens-capped pairs (signal = dark current), in the read-noise regime below the knee.

OP4 Color



Per-patch CIEDE2000 before and after the 3x3 matrix, incandescent light, chart 1 of 3 (legend means are this chart's; the datasheet values are the mean over three charts).